

## Assignment 4

**Date: 8th January, 2026**

**Instructions:** Perform a complete dry run of the program on paper, take clear pictures of the working steps, and compile them into a PDF file for assignment submission.

### Q#01

Find an optimal multiplication order for the following matrices.

$$\begin{matrix} A_1 & \cdot & A_2 & \cdot & A_3 & \cdot & A_4 & \cdot & A_5 \\ (5 \times 4) & & (4 \times 6) & & (6 \times 2) & & (2 \times 7) & & (7 \times 3) \end{matrix}$$

### Q#02

You are given  $n$  activities with **start times** and **finish times**.

Select the **maximum number of activities** that **do not overlap**, assuming **only one activity can be done at a time**..

### Example Input

| Activity | Start | Finish |
|----------|-------|--------|
| A1       | 1     | 2      |
| A2       | 3     | 4      |
| A3       | 0     | 6      |
| A4       | 5     | 7      |
| A5       | 8     | 9      |
| A6       | 5     | 9      |